E-COMMERCE SALES ANALYSIS USING SQL

A Data Analysis Project

Tools & Technologies:
SQL | MySQL | Data Analysis

Dataset: E-Commerce / Blinkit Product Sales Dataset

Executive Summary

This project analyzes retail product and sales data using SQL to understand business performance across products, categories, brands, sellers, locations, pricing, inventory, and delivery.

The analysis focuses on identifying the products and categories driving revenue, understanding demand and sales patterns, evaluating inventory risks, and measuring operational performance.

The goal is not only to calculate metrics, but to translate raw transactional data into insights that can support better **inventory planning, product strategy, seller evaluation, and operational decision-making**.

1. OBJECTIVE

The main objective of this project is to analyze e-commerce product sales data using SQL and convert raw data into meaningful business insights.

The analysis focuses on:

- Identifying the top-selling products based on quantity sold.
- Analyzing total revenue across categories, brands, cities, and sellers.
- Identifying products that require restocking based on stock and reorder levels.
- Analyzing product ratings and customer reviews.
- Understanding the impact of discounts on product sales.
- Identifying high-demand products with low inventory.
- Evaluating seller performance based on sales and revenue.
- Analyzing delivery performance and delivery status.
- Using SQL window functions to rank products within categories.

The overall goal is to demonstrate how SQL can be used to solve real-world business problems and support data-driven decision-making.

2. ABOUT THE DATASET

This project uses an e-commerce product sales dataset containing information about products, pricing, sales, inventory, ratings, sellers, demand, and delivery.

Important columns include:

- product_id – Unique ID of each product.
- product_name – Name of the product.
- category – Product category.
- brand – Product brand.
- price – Original product price.
- discount_pct – Discount percentage.
- final_price – Final selling price after discount.
- rating – Product rating.
- num_reviews – Number of product reviews.
- delivery_time_min – Delivery time in minutes.
- city – City where the product is sold/delivered.
- seller – Seller associated with the product.
- stock – Current stock quantity.
- sold_quantity – Quantity sold.
- profit_margin_pct – Profit margin percentage.
- is_organic – Indicates whether the product is organic.
- packaging_type – Type of packaging.
- shelf_life_days – Product shelf life.
- reorder_level – Stock level at which reordering is required.
- demand_index – Demand level for the product.
- offer_type – Type of offer applied.
- delivery_status – Delivery status

ANALYTIC QUESTIONS

1. Find the Top 10 Best-Selling Products based on sold_quantity.

```sql
SELECT
    product_id, product_name, sold_quantity
FROM
    blinkit.blinkit_dataset
ORDER BY sold_quantity DESC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

product_id	product_name	sold_quantity
4582	Parle Premium Snacks 746	720
5750	Coca-Cola Original Beverages 520	710
6552	DailyGreens Original Fruits 767	705
9557	Dove Family Pack Personal 338	704
2149	Britannia Original Bakery 113	694
5166	P&G Family Pack Personal 364	679
4783	BakeHouse Lite Bakery 360	671
6642	Himalaya Classic Personal 339	670
11535	Patanjali Family Pack Grocery 259	666
9220	Nivea Family Pack Personal 797	656

2. Calculate the Total Revenue for each category.

```sql
SELECT category,
    ROUND(SUM(sold_quantity * price), 2) AS total_revenue
FROM
    blinkit.blinkit_dataset
group by category
order by total_revenue desc;
```

Result Grid | Filter Rows:

	category	total_revenue
▶	Household	133235167.22
	Personal Care	111152222.78
	Grocery	70479311.21
	Bakery	70219008.01
	Fruits & Vegetables	52930484.88
	Beverages	46729078.78
	Snacks	41245046.47
	Dairy	39354324.3

3. Find the average rating of each brand and display only brands with an average rating greater then 4.0.

```
SELECT
    brand, AVG(rating) AS avg_rating
FROM
    blinkit.blinkit_dataset
GROUP BY brand
HAVING AVG(rating) > 4.0;
```

Result Grid | Filter Rows:

	category	total_revenue
▶	Household	133235167.22
	Personal Care	111152222.78
	Grocery	70479311.21
	Bakery	70219008.01
	Fruits & Vegetables	52930484.88
	Beverages	46729078.78
	Snacks	41245046.47
	Dairy	39354324.3

4. Retrive all product where stock is less then the record level.

```sql
SELECT
    product_name, stock, reorder_level
FROM
    blinkit.blinkit_dataset
WHERE
    stock > reorder_level;
```

Result Grid | Filter Rows: | Export:

product_name	stock	reorder_level
Tata Organic Grocery 300	76	15
Mother Dairy Lite Dairy 275	122	24
P&G Classic Personal 439	126	25
Dettol Fresh Household 771	92	18
Minute Maid Daily Beverages 264	152	30
Britannia Premium Snacks 284	114	22
Britannia Organic Bakery 192	99	19
Britannia Premium Bakery 931	96	19
Surf Excel Classic Household 748	101	20
Aashirvaad Fresh Grocery 206	138	27
Harpic Premium Household 296	124	24

...

5. display the top 10 product with the highest profit margin percentage.

```sql
SELECT
    product_name, profit_margin_pct
FROM
    blinkit.blinkit_dataset
ORDER BY profit_margin_pct
LIMIT 10;
```

	product_name	profit_margin_pct
▶	ITC Original Snacks 225	5
	FreshFarm Family Pack Fruits 99	5
	FreshFarm Original Fruits 992	5
	BakeHouse Organic Bakery 919	5
	Gowardhan Daily Dairy 753	5
	FreshFarm Lite Fruits 993	5
	Aavin Organic Dairy 759	5
	Dettol Premium Household 681	5
	Modern Classic Bakery 452	5
	Surf Excel Premium Household 499	5

6. Find the total number of organic product ,there average rating, and total quantity sold.

```
SELECT
    COUNT(*) AS total_organic_product,
    ROUND(AVG(rating), 2) AS average_rating,
    SUM(sold_quantity) AS total_quantity_sold
FROM
    blinkit.blinkit_dataset
WHERE
    is_organic = 'true';
```

	total_organic_product	average_rating	total_quantity_sold
▶	3168	4.19	508591

7. Calculate the total sales revenue for each city.

```
SELECT
    city, SUM(price * sold_quantity) AS total_sale_revenue
FROM
    blinkit.blinkit_dataset
GROUP BY city
ORDER BY total_sale_revenue DESC;
```

city	total_sale_revenue
Pune	59678761.71999999
Lucknow	58728685.349999994
Mumbai	58415002.06999991
Bengaluru	57419257.18999999
Ahmedabad	57411223.53000008
Jaipur	56362204.92000004
Delhi	56266131.669999994
Kolkata	55469519.89000002
Hyderabad	54595138.65000002
Chennai	50998718.66

8. Find each seller's:

- **total number of product**
- **total quantity sold**
- **total revenue**

```
SELECT
    seller,
    COUNT(product_name) AS total_product,
    SUM(sold_quantity) AS total_quantity,
    SUM(final_price * sold_quantity) AS total_revenue
FROM
    blinkit.blinkit_dataset
GROUP BY seller
ORDER BY total_product DESC;
```

seller	total_product	total_quantity	total_revenue
DailyNeeds	2217	354369	86819816.15999998
SellerA	2212	356170	85689834.24999999
SellerB	2175	345970	82414065.1600001
LocalMart	2154	350206	85494650.28999996
QuickStores	2148	354800	83958795.58000001
UrbanSeller	2094	349686	84649969.00000006

8. Calculate the percentage of on-time and delayed delivery based on delivery_status.

```sql
SELECT
    ROUND(SUM(CASE
                WHEN delivery_status = 'on-time' THEN 1
                ELSE 0
            END) * 100.0 / COUNT(*),
            2) AS on_time_percentage,
    ROUND(SUM(CASE
                WHEN delivery_status = 'delayed' THEN 1
                ELSE 0
            END) * 100.0 / COUNT(*),
            2) AS delayed_percentage
FROM
    blinkit.blinkit_dataset;
```

Result Grid | Filter Rows: | Export:

on_time_percentage	delayed_percentage
79.96	20.04

9. Find the top 5 highest rating product with more than 100 reviews.

```sql
SELECT
    product_name, rating, num_reviews
FROM
    blinkit.blinkit_dataset
WHERE
    num_reviews > 100
ORDER BY rating DESC , num_reviews DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export:

product_name	rating	num_reviews
Minute Maid Fresh Beverages 549	5	839
Lizol Organic Household 372	5	823
Aashirvaad Organic Grocery 263	5	765
Minute Maid Classic Beverages 827	5	726
Modern Organic Bakery 583	5	721

10. Compare the average sold quantity of product with discount and without discount.

```sql
SELECT
    CASE
        WHEN discount_pct > 0 THEN 'discount'
        ELSE 'not discount'
    END AS discount_status,
    ROUND(AVG(sold_quantity), 2) AS avg_quantity
FROM
    blinkit.blinkit_dataset
GROUP BY discount_pct;
```

discount_status	avg_quantity
discount	157.37
discount	173.21
not discount	162.47
discount	168.79
discount	158.62
discount	161.06
discount	161.27

11. Find the average rating and total revenue for each packaging_type.

```sql
SELECT
    packaging_type,
    ROUND(AVG(rating), 2) AS avg_rating,
    ROUND(SUM(final_price * sold_quantity), 2) AS total_revenue
FROM
    blinkit.blinkit_dataset
GROUP BY packaging_type;
```

packaging_type	avg_rating	total_revenue
Can	4.19	85053354.81
Jar	4.19	84158933.86
Bottle	4.2	90507323.06
Packet	4.2	82373684.46
Pouch	4.21	81709651.01
Box	4.19	85224183.24

12. Display product with a shelf life greatest then 180 days and a rating above 4.5.

```sql
SELECT
    product_name, shelf_life_days, rating
FROM
    blinkit.blinkit_dataset
WHERE
    shelf_life_days > 180 AND rating > 4.5;
```

product_name	shelf_life_days	rating
Lay's Classic Snacks 961	193	5
Lizol Original Household 737	1637	4.6
Patanjali Daily Grocery 796	440	5
Lizol Family Pack Household 496	864	4.8
Britannia Daily Snacks 534	198	5
Surf Excel Original Household 330	575	5
Lay's Classic Snacks 607	364	5
Parle Organic Snacks 640	311	5
Patanjali Classic Grocery 588	451	4.6
P&G Lite Personal 332	863	4.7
Dove Premium Personal 235	1135	4.8

…

13. Find product where:

- **Demand_index > 80**
- **Stock > reorder_level**
- **these product require immediate restoring.**

```sql
SELECT
    product_name, demand_index, stock, reorder_level
FROM
    blinkit.blinkit_dataset
WHERE
    demand_index > 80
        AND stock > reorder_level;
```

product_name	demand_index	stock	reorder_level
P&G Classic Personal 439	100	126	25
Britannia Family Pack Bakery 876	100	92	18
Britannia Premium Bakery 316	97	97	19
P&G Lite Personal 892	90	112	22
P&G Family Pack Personal 921	92	104	20
Parle Fresh Snacks 787	94	131	26
Parle Organic Snacks 640	100	113	22
Lizol Family Pack Household 174	100	129	25
Pepsi Original Beverages 130	86	135	27
ITC Fresh Snacks 386	90	91	18
Dove Premium Personal 235	100	89	17
FreshFarm Classic Fruits 367	88	130	26

…

14. Rank product within each category based on revenue. (using a window function rank(),dense_rank()).

```sql
select category,product_name,final_price*sold_quantity as revenue,
rank()over(partition by category
            order by final_price*sold_quantity desc) as revenue_rank
from blinkit.blinkit_dataset;
```

category	product_name	revenue	revenue_rank
Bakery	Britannia Organic Bakery 134	259803.18	1
Bakery	Modern Daily Bakery 700	247614.44	2
Bakery	BakeHouse Fresh Bakery 743	245336.34	3
Bakery	Modern Premium Bakery 876	237012.30000000002	4
Bakery	Modern Family Pack Bakery 478	228285	5
Bakery	Britannia Original Bakery 537	211025.02000000002	6
Bakery	Britannia Classic Bakery 823	203834.61000000002	7
Bakery	Modern Lite Bakery 151	202302.9	8
Bakery	Britannia Family Pack Bakery 876	202175.1	9
Bakery	Britannia Original Bakery 534	194859.38999999998	10
Bakery	BakeHouse Organic Bakery 256	184615.59	11
Bakery	Britannia Lite Bakery 275	184421.6	12
Bakery	Modern Daily Bakery 341	183678.38	13

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15. Find the top 5 categories by total revenue.

```sql
SELECT
    category, final_price * sold_quantity AS total_revenue
FROM
    blinkit.blinkit_dataset
LIMIT 5;
```

category	total_revenue
Grocery	36111.44
Dairy	868.56
Personal Care	292158.79
Household	20696.609999999997
Beverages	4149.12

16. Calculate the average delivery time for each city.

```sql
SELECT
    city, AVG(delivery_time_min) AS avg_delivery_time
FROM
    blinkit.blinkit_dataset
GROUP BY city
ORDER BY avg_delivery_time DESC;
```

Result Grid | Filter Rows:

city	avg_delivery_time
Lucknow	34.4977
Jaipur	32.7120
Ahmedabad	29.5749
Mumbai	29.3781
Bengaluru	27.4558
Kolkata	26.6209
Chennai	25.6158
Delhi	24.3888
Hyderabad	23.4257
Pune	21.5393

17. Find the most expensive product in each category.

```sql
select product_name,category,final_price
        from (select *,
    rank()over( partition by category
order by final_price desc) as rnk
from blinkit.blinkit_dataset) as t
where rnk=1;
```

Result Grid | Filter Rows: | Export: | W

product_name	category	final_price
BakeHouse Premium Bakery 887	Bakery	498.78
Coca-Cola Organic Beverages 748	Beverages	348.59
Gowardhan Premium Dairy 333	Dairy	249.19
DailyGreens Organic Fruits 370	Fruits & Vegetables	399.81
Patanjali Daily Grocery 375	Grocery	499.77
Surf Excel Lite Household 114	Household	998.92
Dove Classic Personal 445	Personal Care	799.55
Parle Family Pack Snacks 693	Snacks	299.89

18. Find the top 3 brand with highest total sales revenue.

```sql
SELECT
    brand,
    SUM(final_price * sold_quantity) AS total_sales_revenue
FROM
    blinkit.blinkit_dataset
GROUP BY brand
ORDER BY total_sales_revenue DESC
LIMIT 3;
```

Result Grid | Filter Rows:

brand	total_sales_revenue
Britannia	31547894.039999973
Harpic	31508881.000000022
Lizol	31077392.449999988

19. Create a sales dashboard query showing:

```sql
SELECT
    COUNT(DISTINCT product_name) AS total_product,
    SUM(final_price * sold_quantity) AS total_revenue,
    SUM(sold_quantity) AS total_sold_quantity,
    ROUND(AVG(rating), 2) AS average_rating,
    (SELECT
            product_name
        FROM
            blinkit.blinkit_dataset
        ORDER BY sold_quantity DESC
        LIMIT 1) AS best_selling_product,
    (SELECT
            category
        FROM
            blinkit.blinkit_dataset
        GROUP BY category
        ORDER BY SUM(sold_quantity * final_price) DESC
        LIMIT 1) AS highest_revenue_category,
    (SELECT
            seller
        FROM
            blinkit.blinkit_dataset
        GROUP BY seller
        ORDER BY SUM(sold_quantity * final_price) DESC
        LIMIT 1) AS best_seller_revenue,
    ROUND(SUM(CASE
                WHEN delivery_status = 'on-time' THEN 1
                ELSE 0
            END) * 100.0 / COUNT(*),
            2) AS on_time_percentage
FROM
    blinkit.blinkit_dataset;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_product	total_revenue	total_sold_quantity	average_rating	best_selling_product	highest_revenue_category	best_seller_revenue	on_time_percentage
12617	509027130.44000053	2111201	4.2	Parle Premium Snacks 746	Household	DailyNeeds	79.96